



COURSE NAME : Design & Analysis of Algorithms
COMPONENT : Midsem (Closed Book)
DATE/DAY : 8th April 2025, Tuesday

COURSE CODE : CS F364
WEIGHTAGE : 30%
DURATION : 90 mins

PART – A (15 Marks)

Note: Show the steps clearly to score full marks

1

	c) Calculate the middle term, which accounts for the cross-product of parts not yet included in the previous two. All computations should be shown in binary.	[2]
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	d) Compute the final product $X.Y$. You can combine all parts to get the final product. Give your answer in binary.	[2]
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2.	Apply the below Quick Sort algorithm on the following array: A=[20, 12, 35, 10, 40, 30, 25] QuickSort(A, low, high) if low < high: pivotIndex = Partition(A, low, high) QuickSort(A, low, pivotIndex - 1) QuickSort(A, pivotIndex + 1, high) pivot = A[high] i = low - 1 Partition(A, low, high) for j = low to high - 1: if A[j] ≤ pivot: i = i + 1 swap A[i] and A[j] swap A[i + 1] and A[high] return i + 1 a) Using the pivot as the last element in each recursive call, fill in the steps below for the first partitioning round only and indicate the final position of the pivot (25) after the partition? First row is filled for your reference. <table><tr><th>Step</th><th>j</th><th>A[j]</th><th>Compare A[j] ≤ pivot?</th><th>Action Taken</th><th>i Value</th><th>Array After Step</th></tr><tr><td>1</td><td>0</td><td>20</td><td>Yes</td><td>Swap (no change)</td><td>0</td><td>[20, 12, 35, 10, 40, 30, 25] (no change)</td></tr><tr><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Step	j	A[j]	Compare A[j] ≤ pivot?	Action Taken	i Value	Array After Step	1	0	20	Yes	Swap (no change)	0	[20, 12, 35, 10, 40, 30, 25] (no change)	2							[3]
Step	j	A[j]	Compare A[j] ≤ pivot?	Action Taken	i Value	Array After Step																	
1	0	20	Yes	Swap (no change)	0	[20, 12, 35, 10, 40, 30, 25] (no change)																	
2																							

3						
4						
5						
6						
7	-	-	-	Swap		Final array after partition:

- b) Fill in the recursive calls and subarrays that Quick Sort will process. Use partitioning where the pivot is always the last element of the current subarray. At each step, fill in the subarray before partitioning, the pivot used, the result after partition, and the two recursive subarrays.

Note:

- i) For each recursive call, fill the following in the table:

- The subarray before partition
- The pivot chosen (last element)
- The subarray after partitioning
- The left subarray ($\leq$ pivot)
- The right subarray ($>$ pivot)

- ii) Include **base cases** (subarrays of size 1 or 0), even though no partition happens — just write "**base case – no partition**" in the last three columns.

- iii) To help track recursion paths, label each call as:

- 0 = Initial (first call)
- 1L = Left subarray from level 0

[4]

- 1R = Right subarray from level 0
- 2L = Left subarray from level 1L, and so on
- This shows which recursive call led to which (like a tree)

Recursion Level	Subarray Before Partition	Pivot Chosen (Last Element)	Subarray After Partition	Left Subarray ($\leq$ pivot)	Right Subarray ($>$ pivot)
0 (Initial)	[20, 12, 35, 10, 40, 30, 25]	25			
1 (Left of Level 0)					
2 (Left of Level 1)					
2 (Right of Level 1)					
3 (Left of Level 2R)					
3 (Right of Level 2R)					
1 (Right of Level 0)					
2 (Left of Level 1R)					
2 (Right of Level 1R)					

***** ALL THE BEST *****



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI
SECOND SEMESTER 2024 – 2025
DEPARTMENT OF COMPUTER SCIENCE

R

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DURATION : 90 mins

Answer all the Questions

PART – B (15 Marks)

1. A thief wants to maximize the total profit while robbing a store, but he can carry a maximum weight of 15 units in his knapsack. He finds 7 objects, each having a specific profit and weight, as shown in the table below:

Objects	O1	O2	O3	O4	O5	O6	O7
Profits	6	10	18	15	3	5	7
Weight	1	2	4	5	1	3	7

Using the Fractional Knapsack approach, answer the following:

- Which objects will be placed inside the knapsack to maximize the total profit?
- What fraction of each selected object will be included in terms of weight?
- What will be the total profit after placing these objects optimally in the knapsack?
- What is the time complexity for solving this particular question using the Fractional Knapsack approach? Justify Your Answer

[1]
[1]
[2]
[2]

2.	Consider the following jobs, each having a specific profit and deadline, as shown in the table below: <table><tr><td>Jobs</td><td>J1</td><td>J2</td><td>J3</td><td>J4</td><td>J5</td><td>J6</td><td>J7</td></tr><tr><td>Profits</td><td>250</td><td>230</td><td>240</td><td>350</td><td>170</td><td>150</td><td>100</td></tr><tr><td>Deadlines</td><td>6</td><td>4</td><td>4</td><td>3</td><td>5</td><td>3</td><td>2</td></tr></table> Using the job sequencing with deadlines approach, answer the following: (a) If each job takes 1 unit of time to complete and the timeline is of 6 units (maximum deadline), which jobs should be scheduled to maximize the profit? (b) What is the correct order of job scheduling? Represent it in a timeline (c) What is the total profit earned from the scheduled jobs?	Jobs	J1	J2	J3	J4	J5	J6	J7	Profits	250	230	240	350	170	150	100	Deadlines	6	4	4	3	5	3	2	[1] [2] [1]
Jobs	J1	J2	J3	J4	J5	J6	J7																			
Profits	250	230	240	350	170	150	100																			
Deadlines	6	4	4	3	5	3	2																			

3.	Kruskal's algorithm uses_____and prim's algorithm uses ____ in determining the MST - edges, vertex - vertex,edges - edges,edges - vertex,vertex	[1]
4.	Dijkstra's Algorithm is used to solve _____ problems. - All pair shortest path - Single source shortest path - Finding minimum cost spanning tree - Greedy algorithms	[1]
5.	Which of the following is the most commonly used data structure for implementing Dijkstra's Algorithm? - Queue - Stack - Graph - Min priority queue	[1]
6.	Consider a undirected graph G with vertices $\{A, B, C, D, E\}$. In graph G, every edge has distinct weight. Edge CD is edge with minimum weight and edge AB is edge with maximum weight. Then, which of the following is false? - Every minimum spanning tree of G must contain CD - If AB is in a minimum spanning tree, then its removal must disconnect G - No minimum spanning tree contains AB	[1]

	d. G has a unique minimum spanning tree	
7.	What is the space complexity of Dijkstra algorithm if V is the number of vertices and E is the number of edges? a. $O(V + E)$ b. $O(V + E \log E)$ c. $O(V \log E)$ d. $O(E \log E)$	[1]

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C&D

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Answer all the Questions

PART – C (15 Marks)

1. Consider the recurrence relation:

$$T(n) = \begin{cases} 2T(n^{1/3}) + O(n), & n > 1 \\ 1, & n = 1 \end{cases}$$

Solve the recurrence using the changing variable method and answer the following:

- (a) Define a suitable change of variable (let's say m) to simplify the recurrence. [1]
- (b) Express $T(n)$ in terms of the new variable m (i.e., define $S(m)$). [1]
- (c) Solve the recurrence $S(m)$ to determine complexity. [2]
- (d) Convert the solution of $S(m)$ back to $T(n)$ and determine the overall time complexity? [1]

2. Consider the recurrence relation:

$$T(n) = \begin{cases} T(n-1) + n, & n > 1 \\ 1, & n = 1 \end{cases}$$

Solve the recurrence using the iteration method:

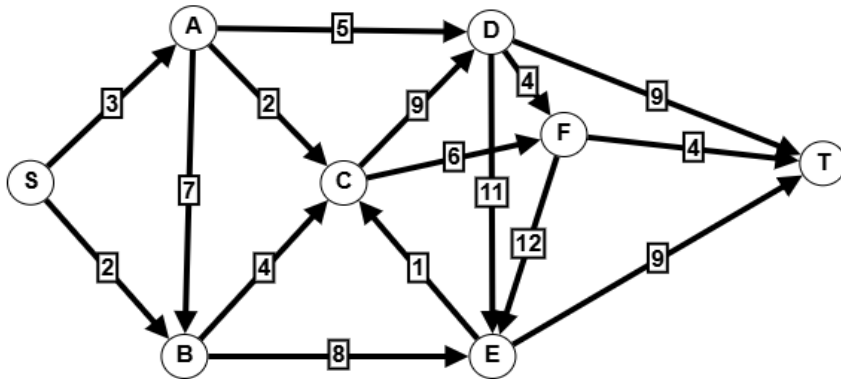
- (a) Expand the recurrence for the first four values of n . [1]
- (b) Identify the pattern and express $T(n)$ in terms of summation. [2]
- (c) Solve the summation to determine the worst-case time complexity of $T(n)$? [2]

3. Given the letters a, b, c, d, e, and f with probabilities of their occurrence 1/3, 1/9, 1/27, 1/81, 1/243, and 1/243, answer the following questions based on Huffman encoding:

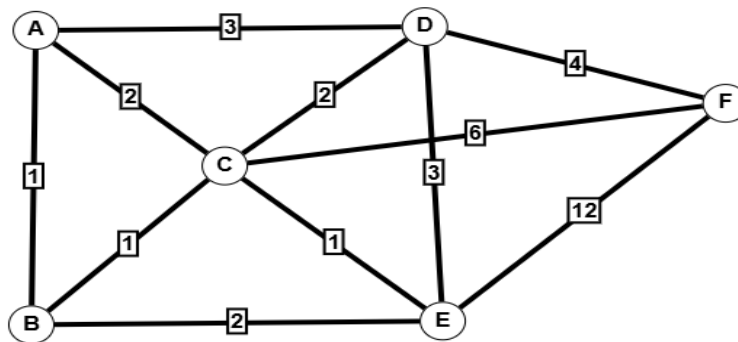
- (a) Construct the Huffman tree for encoding these letters? [3]
- (b) Identify the correct Huffman code for the given letters from the options below? [2]
 - i) 1, 01, 001, 0001, 00000, 00001
 - ii) 11, 10, 011, 010, 001, 000
 - iii) 11, 10, 01, 001, 0001, 0000
 - iv) 110, 100, 010, 000, 001, 111

PART – D (15 Marks)

1. John is driving his electric car to go from Dubai (i.e., source S) to Abu Dhabi (i.e., destination T). Due to the limitations of charging on the road, he needs to find the shortest path. There are multiple paths that exist from Dubai (S) to Abu Dhabi (T) and the shortest path requires less time and battery power. The below graph $G = (V, E)$ presents the roadmap between Dubai and Abu Dhabi. The edge weights W represent the distance.
 - (a) Find the shortest path from source S (i.e., Dubai) to destination T (i.e., Abu Dhabi) using the Dijkstra algorithm and show all the required steps. [5]
 - (b) Draw the shortest path from S to T and compute the shortest distance. [2]



2. Apply Prim's and Kruskal's algorithms separately on the below graph and find the minimum spanning tree. Show step-by-step procedure for both the algorithms to find the minimum spanning tree. Illustrate a scenario in which both algorithms may find different minimum spanning trees. [8]



***** ALL THE BEST *****



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SECOND SEMESTER 2024 – 2025
DEPARTMENT OF COMPUTER SCIENCE

A

COURSE NAME : Design & Analysis of Algorithms

COMPONENT : Compre (Closed Book)

DATE/DAY : 29th May 2025, Thursday

COURSE CODE : CS F364

WEIGHTAGE : 40%

DURATION : 180 mins

Answer all the Questions

SECTION – A (1 X 10 = 10 Marks)

Write True or False for each of the following statements to indicate whether the statement is true or false, respectively, and briefly explain why (one or two sentences). Your justification is worth more points than your true-or-false designation. Careful: Some problems are straightforward, but some are tricky!

1. Suppose that every operation on a data structure runs in $O(1)$ amortized time. Then the running time for performing a sequence of n operations on an initially empty data structure is $O(n)$ in the worst case.
2. Using the substitution method to solve the recurrence $T(n) = \sqrt{n}T(\sqrt{n}) + n$ will give the solution as $T(n) = O(n \lg \lg n)$.
3. If a problem has an algorithm that is correct for $2/3$ fraction of the inputs, then it also has an algorithm that is correct for 99.9% of the inputs.
4. There is an NP-hard problem with a known polynomial time randomized algorithm that returns the correct answer with probability $2/3$ on all inputs.
5. Every special case of an NP-hard problem is also NP-hard.
6. If Vertex Cover reduces to Set Cover in polynomial time, and Set Cover is in P, then Set Cover is NP-complete.
7. If we use a max-queue instead of a min-queue in Kruskal's MST algorithm, it will return the spanning tree of maximum total cost (instead of returning the spanning tree of minimum total cost). (Assume the input is a weighted connected undirected graph.)
8. If a problem P has a polynomial-time approximation algorithm with approximation ratio 1, then P is NP-complete.
9. If a Linear Programming problem has a unique optimal solution, then it must occur at a vertex (corner point) of the feasible region.
10. Increasing the number of processors in a parallel program always decreases the total execution time.

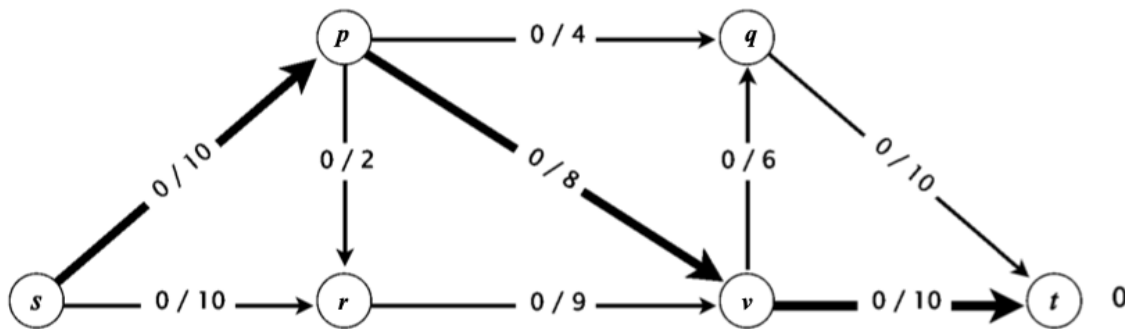
SECTION – B (5 X 3 = 15 Marks)

11. You're given jobs with identical deadlines but varying profits, and only one can be selected per unit time.

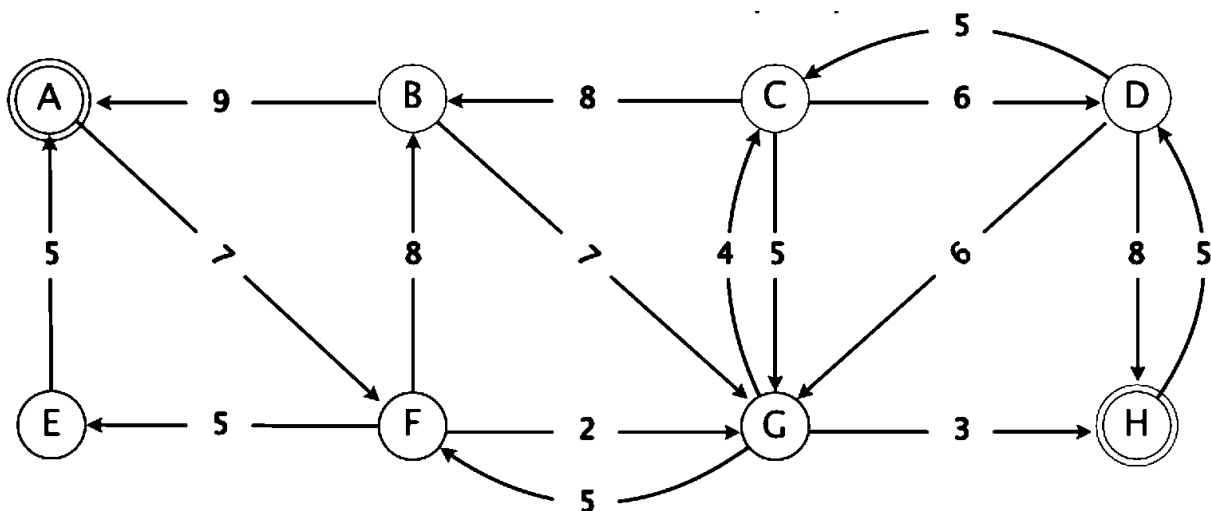
Job	Deadline	Profit
A	2	10
B	2	20
C	2	30

Show how greedy selection by profit works well here but construct a variation where it fails to find the optimal set. Justify with an alternate job list (invent values).

12. Consider the given network flow and the path chosen p . Show the augment flow along path P and the cost.

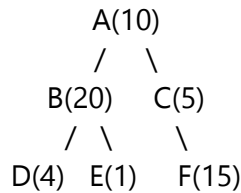


13. For the given graph, find the augmenting path of highest bottleneck capacity? Also mention the highest bottleneck capacity.



14. The Ford-Fulkerson algorithm finds augmenting paths in a residual graph and updates flows iteratively. Suppose you attempt to parallelize it by searching for multiple disjoint augmenting paths simultaneously in each iteration and updating flows accordingly.
- Is it possible to solve the maximum flow problem efficiently using parallel programming in general? Justify with one practical limitation.
 - What would be the parallel time complexity per iteration (assuming k disjoint paths are found in parallel)?
15. You are given a rooted binary tree. Each node has a weight as shown. You are allowed to cut exactly one edge, breaking the tree into two components. Your goal is to minimize the weight of the heavier component.
- Draw the tree using the node weights below.

- (b) Identify which edge you would cut to minimize the maximum weight of the resulting two components.
- (c) Briefly justify your choice.



You may assume the root is A. All node weights are in parentheses. You must consider all possible edge cuts.

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R

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WEIGHTAGE : 40%

DURATION : 180 mins

Answer all the Questions

SECTION – C (5 X 5 =25 Marks)

16. A factory manufactures two milk products A and B . To manufacture one unit of A , 1.5 kg sugar and 2.5 litres of milk are required. To manufacture product B , 2.5 kg sugar and 1.5 litre of milk are required. The manufacturing unit has 300 kg of sugar and 240 litres of milk. Profit per unit for product A is Rs. 50 and for B is Rs. 40. Formulate this problem using linear programming and solve it using simplex method.
17. The N -Queens Problem involves placing N queens on an $N \times N$ chessboard such that no two queens threaten each other. It means no two queens share the same row, column, or diagonal. This traditional problem is modified and added an additional constrain that is the first queen ($Q1$) must be placed in column 3 of the first row. Solve this modified problem using the backtracking approach for $N = 4$, and find all valid solutions that satisfy the given constraint. Also, draw the corresponding state space tree to illustrate the search process, and explain how backtracking helps to improve performance.
18. Consider the following recursive algorithm
 Algorithm TrickyAlgo(n):
 if $n \leq 1$:
 return 1

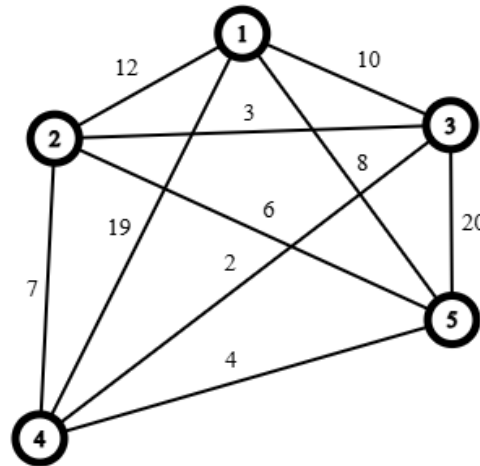
```

for i = 1 to n2:
    do constant time work
return TrickyAlgo(n / 2) + TrickyAlgo(n / 2) + TrickyAlgo(n / 2)

```

Use the Master Theorem to find the time complexity of this algorithm. Show your working and reasoning clearly.

19. The Traveling Salesman Problem (TSP) is an optimization problem that aims to find the shortest possible route that visits each city exactly once and returns to the starting city. Since TSP is an NP-Hard problem, it becomes computationally infeasible to solve exactly for large input sizes. Suggest a suitable algorithm to solve this problem in linear or polynomial time to find near optimal solution. Apply your algorithm step-by-step on the below graph and demonstrate how it works and find the solution. Write down the time complexity of your algorithm. [2+2+1]



20. You are given an array `arr[]` of length `n` representing the height of `n` buildings. A frog is initially on the first building and wants to reach the last building. The frog can jump either to the next building or skip one building. The cost to jump from building `i` to `j` is $\text{abs}(\text{arr}[i] - \text{arr}[j])$. Write a Dynamic Programming solution to find the minimum total cost for the frog to reach the last building.

SECTION – D (3 X 10 = 30 Marks)

21. Consider the Fractional Knapsack problem where knapsack capacity is 13 kg and available items with their profit and weight are given below. Solve the following question.

Item	X1	X2	X3	X4
Profit	7	9	10	12
Weight (in kg)	4	6	7	10

- Formulate the above problem using standard linear programming (LP). [1]
 - Show the LP formulation using a matrix after introducing required slack variables. [1]
 - Solve the problem using simplex method and answer to the following questions. [6+2]
 - Selected pivot element in each iteration of the method.
 - Order of the selected items by the simplex method.
 - Total profit gain found by the method.
 - Can we solve 0/1 knapsack using LP? Justify.
22. RoboTask Inc., a robotics company, has received five specialized projects (J1 to J5) that must be handled by five of their most skilled technicians (W1 to W5). The company wants to ensure each technician is assigned exactly one unique project, and each project gets exactly one technician.

Each technician has a specific cost (in effort units) associated with handling each project, based on their expertise and the complexity of the task. The cost matrix is as follows:

	J_1	J_2	J_3	J_4	J_5
W_1	9	2	7	8	6
W_2	6	4	3	7	5
W_3	5	8	1	8	3
W_4	7	6	9	4	2
W_5	5	3	5	6	7

However, there are operational constraints due to equipment compatibility, technician preferences, and safety protocols:

1. W_3 refuses to work on J_4 due to a past incident.
2. J_2 can only be handled by either W_1 or W_2 due to its high-security clearance.
3. No technician is allowed to be assigned a job with a cost higher than 8, as it violates the fatigue policy.

The company's goal is to minimize the total assignment cost while respecting all constraints.

Additional Rules for Tree Construction:

- Node exploration rule 1: If two nodes in the search tree have the same lower bound and are at levels i and $i+1$, explore the node at level $i+1$ (farthest from root) first.
- Node exploration rule 2: If two nodes in the search tree have the same lower bound and are at the same level, explore nodes from right to left.

Note: Initially, the current best (minimum) cost is assumed to be infinity.

23. The City Museum is planning a special exhibition, and the security team needs to place guards at certain rooms to ensure all hallways connecting the rooms are under surveillance.

The museum layout is represented as follows:

- There are 6 rooms: A, B, C, D, E, and F.
- The rooms are connected by hallways:
 - Room A is connected to Rooms B and C
 - Room B is connected to Rooms A, C, and D
 - Room C is connected to Rooms A, B, and E
 - Room D is connected to Rooms B and E
 - Room E is connected to Rooms C, D, and F
 - Room F is connected only to Room E

- (a) Model the above scenario as Problem A, where the objective is to assign guards to at most k rooms such that every hallway is watched. A hallway is considered watched if at least one guard is placed at either of its ends. Find the value of k ?

- (b) Reduce Problem A into Problem B, as follows:

- Define the universe U as the set of all hallways.
- Represent each room as a set containing all hallways incident to it.

List the universe U and the sets corresponding to each room?

- (c) Solve Problem B, to determine whether we can select $\leq k$ subsets to cover all hallways in U ?

- (d) Now, consider Problem C, which Ram is analysing to determine its computational complexity. Ram demonstrates that Problem B can be reduced to Problem C in polynomial time. Given that Problem B is known to be NP-Complete. What can you conclude with certainty about the nature of Problem C? Provide a clear and rigorous justification for your answer, referencing the concepts of polynomial-time reductions and NP-completeness.

***** ALL THE BEST *****